



Acceptance Test Plan

Mars Hard Lander Electrical System

WATTS UP

1



Meet the team



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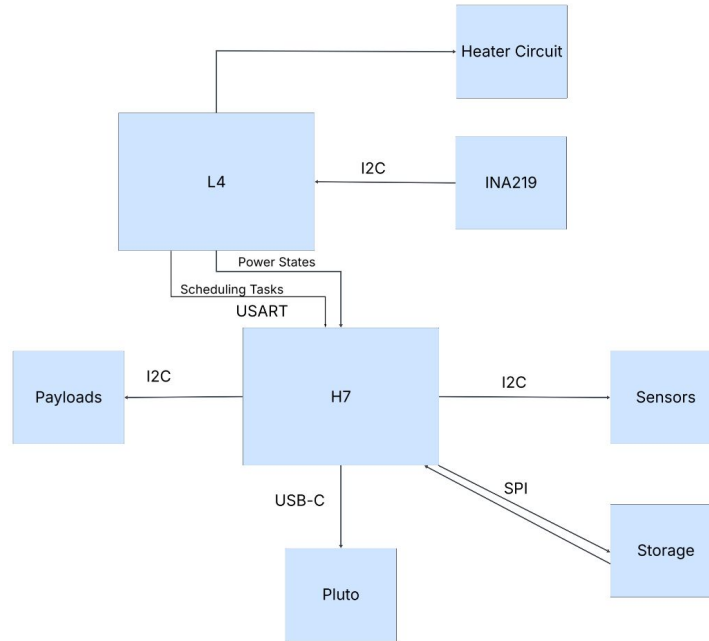
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Test Block Diagram



Marketing Requirements and Use Cases:



- **Marketing Requirement**

- Powered by solar and batteries
- Withstand martian temperatures
- Collect environmental sensor data
 - Temperature
 - Pressure
 - Humidity
 - UV
 - Acceleration
- Transmit data to receiver

- **Use Cases**

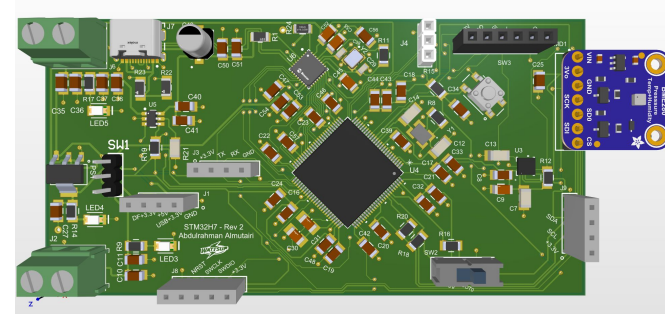
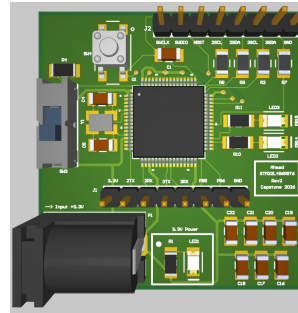
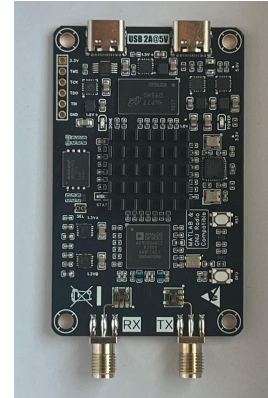
- The lander autonomously collects environmental data and transmits it to receiver
- The electrical system maintains a low-power survival state to preserve functionality through L4 controlling the power state of the H7



Test Procedure



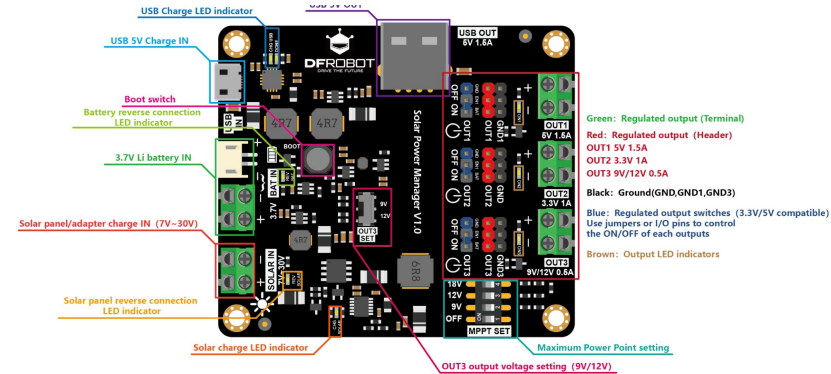
- **Power**
 - Connect Li-Ion battery and solar panel to solar manager
 - Verify regulated power to mainboard and RF subsystem
 - Confirm stable voltage rails before operation
 - Confirm battery is charging
- **Mainboard**
 - Initialize STM32L4 and STM32H7
 - Read sensor data
 - Compare temperature readings for consistency
 - Write sensor data to onboard SPI flash
- **PlutoSDR RF**
 - Retrieve stored data from flash
 - Send data from H7 to PlutoSDR via USB
 - Transmit data over RF to receiver
 - Verify successful reception
- **Software**
 - L4 schedules system operation (5-min cycles)
 - Send command from L4 → H7
 - H7 executes tasks (sense, store, transmit)
 - Return system to low-power state after completion



Power System Test: Backup Plan Power Board

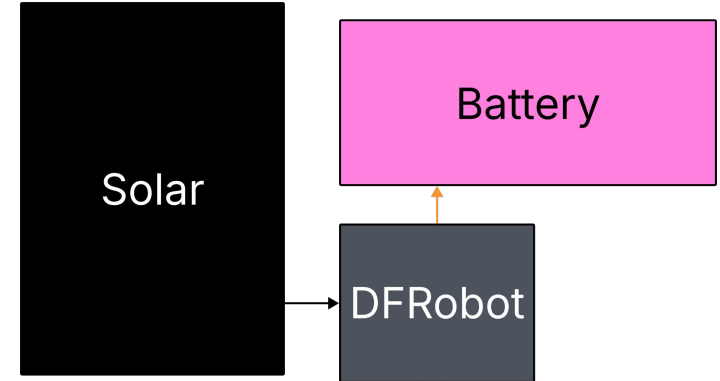
DFR Board (Plan B) Test:

- Solar power in direct light connected to DFRobot Board (Checked Output)
- Battery connected simultaneously
- Battery is being charged by the solar panel while 3.3v, 5v, and 12v are held constant (**Hard to tell**)



Other Tests Performed:

- Cold tested for at 48 hours at -40F
- Battery Discharge tests passed:
 - Charge lasted more than 48 hours under expected load in -40F



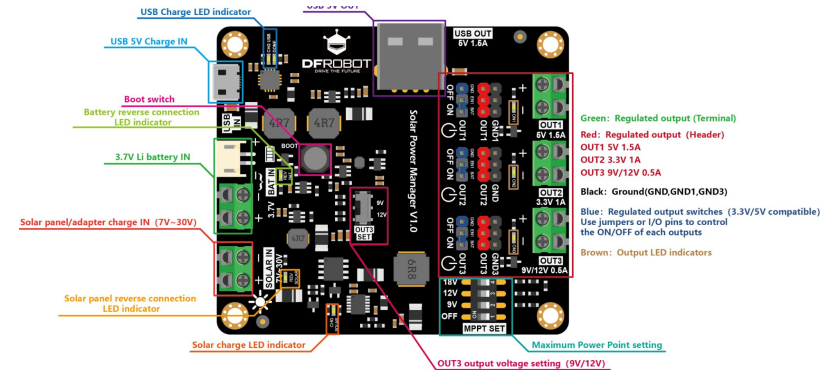
Power Subsystem Purpose & Setup

Custom Battery:

- 1S9P pack connected to DFRobot board to provide power distribution:
 - 3.3v, 5v, 12v
 - Powering mainboard + onboard sensors
 - STM32 L4, H7, and RF Pluto SDR act as a load with sensors connected
 - Arranged flatly to fit in the enclosure



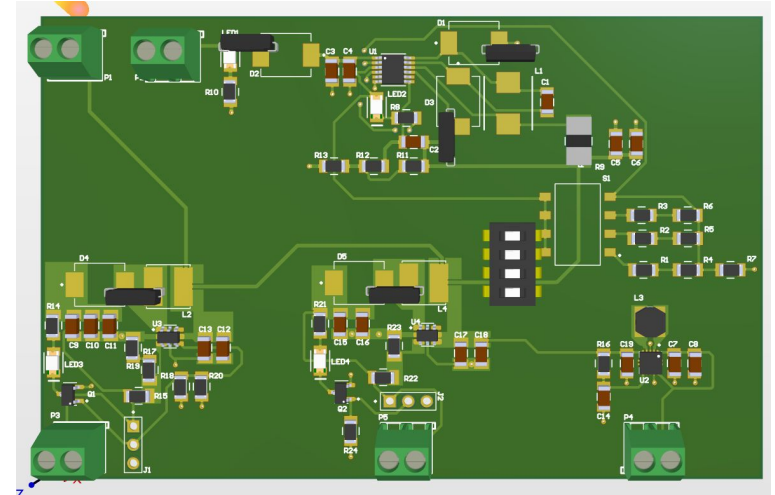
- Solar panel connected to DFRobot to charge battery



Test Setup: Custom Power

Custom Power Board:

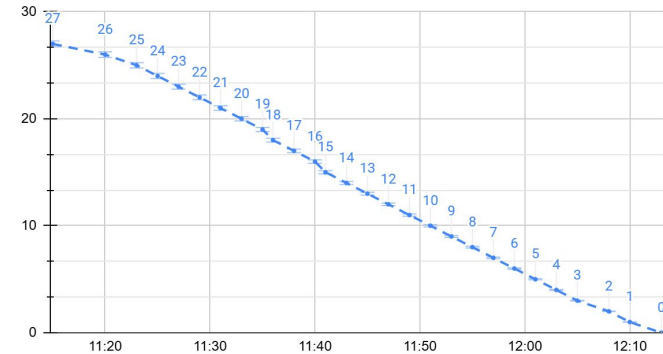
- Delayed Boost Convertors
- 3.3v Buck-Boost currently not working
- **Test Performed:**
 - Input voltage of 4v with a power supply set with current limit of 0.25A
 - Current limiting at 0.25A at the power supply when open load
- **Possible reasons:**
 - Bad routing/floorplanning → Better routing on rev 2
 - Poor Soldering → Stencil for soldering



Enclosure Setup

- We put a thermocouple to read into the enclosure temp and we record the measurements.
- We mark down the time for each drop of one degree celsius.
- The results show that roughly ever minuet the enclosure dropped one degree with only one inner and one outer layer of insulation.
- The plan is to now increase the numer of layers

Enclosure Temp Over Time



Setup: Mainboard

1. The **STM32L4** functions as the supervisory controller. Communicates with the **STM32H7** over a **USART** interface, allowing the L4 to send commands and coordinate system operation.
2. The **STM32H7** functions as the primary processing unit and communication host. It interfaces with the **USB300** high-speed **USB PHY**, enabling the H7 to operate in **USB host mode**.
3. The **Pluto SDR Nano** is connected to the STM32H7 through the USB interface. The STM32H7 provides **5V power to the Pluto SDR** and manages the RF communication pipeline.

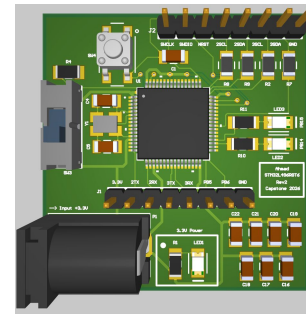


Figure 1: STM32L4

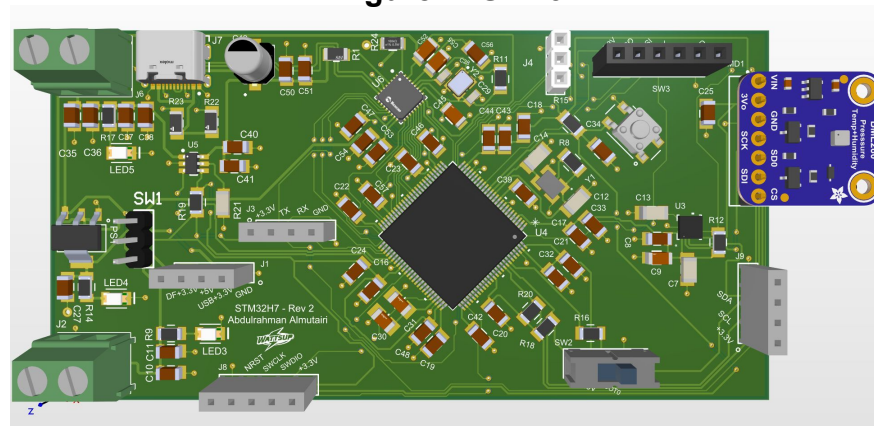
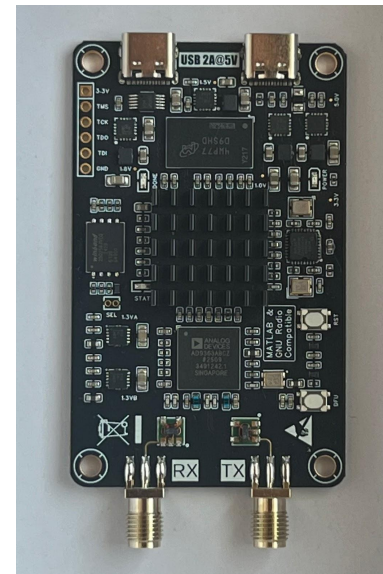


Figure 2: STM32H7



Setup: RF

- PlutoSDR board communicates with receiver using BPSK modulation with RRC filtering and error correction parity bits
- SDR will communicate with Mainboard over high speed USB
- SDR will transmit stored sensor data over RF to receiver



PlutoSDR Nano



Results:



Enclosure

- The enclosure lasted an hour and in that time went from 27C to 0C, this shows promising results for when the inner core is installed to give the internals their own thermal layer to shield them from losses

Solar

- The solar panel charges battery but very low current
- Output voltages remain constant with and without battery



Live demo Results:



- **Thermocouple vs BME688**
 - **Two independent temperature measurements:**
 - Thermocouple
 - BME688 digital sensor
 - Real time readings compared during the live demonstration
- **Observed:**
 - Consistent temperature tracking between both



So What?



Enclosure

- Protects electronics from environmental damage
- Ensure system reliability for long term deployment
- Support mounting

Commands

- L4 → H7 command system enables controlled execution per cycle
- Bit encoded commands allow flexible task selection
- Reduce power consumption

Scheduling + Power States

- L4 timer based scheduling enables autonomous operation
- No human interaction after deployment
- Constant sampling intervals (writing sensor data every 5 mins)
- L4 → H7 activation only when needed
- H7 perform high power tasks, then returns to sleep
- Extend battery life (critical ERD metric)



So What?



Data collection and storage

- Sensor data logged every 5 minutes → continuous dataset
- Onboard storage removes dependency on constant RF connection
- Enables post mission analysis and validation

Data Transmission

- Data stored long-term (1 month)
- Bulk transfer instead of continuous transmission → Energy efficient
- Ensures mission data is preserved until transmission window approx (every 1 month)

Solar Panels/Power System

- Extend our mission duration by enabling renewable energy source to charge our 1S9P Li-Ion battery pack
- Critical for meeting our ERD mission lasts > 2 martian years e.g. 3.75 Earth years



EXPO Day

- Final Rev PCB: **4-layer board (6" × 4")**
- Integrated system on a single board
- Includes:
 - **STM32H7** → high-performance processing
 - **STM32L4** → low-power scheduling & control power states
 - **USB3300 (ULPI)** → High-Speed USB for PlutoSDR
 - **DFRobot Solar Power Manager** → power regulation distribution & battery management

